FINITE LOCAL FREENESS OF A QUASI-COHERENT MODULE DESCENDS ALONG AN FPQC COVERING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- A quasi-coherent sheaf is finite locally free if and only if it is of finite presentation and flat. Algebra, Lemma algebra-lemma-finite-projective, invoked at proof lines 2231-2233. Splits finite local freeness into finite presentation and flatness.
- Flatness and finite presentation of quasi-coherent modules descend along an fpqc covering. Same chapter, Lemmas lemma-flat-descends and lemma-finite-presentation-descends, invoked at proof lines 2234-2237. Gives the two ingredients for F.

Source reference. Original lemma 00NX: sources/stacks-chow/tags.txt, source lines 875–875. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05B0: sources/stacks-algebra/pinned-original-source.tex, source lines 31681–31681. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05B1: sources/stacks-descent/pinned-original-source.tex, source lines 2208–2208. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $\{f_i : X_i \to X\}_{i \in I}$ be an fpqc covering such that each $f_i^*\mathcal{F}$ is a finite locally free $\mathcal{O}_{X_i}$-module. Then $\mathcal{F}$ is a finite locally free $\mathcal{O}_X$-module.*

Proof. This follows from the fact that a quasi-coherent sheaf is finite locally free if and only if it is of finite presentation and flat, see Algebra, Lemma 00NX. Namely,

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if each $f_i^*\mathcal{F}$ is flat and of finite presentation, then so is $\mathcal{F}$ by Lemmas 05B1 and 05B0. $\square$